Notes on sections 5-1 & 5-2

Read pp. 121 - 122, 124, 126 - 129

Pp. 121 - 122 & 124

We have been covering decision theory, which, as you know, concerns a single person making a decision under complete uncertainty or risk. The goal there was to determine what decision a rational person would make using different rules and guidelines.

(The term *strategy* has different meanings in decision theory and in game theory, and so shortly, you will have to start using the game theory meaning.)

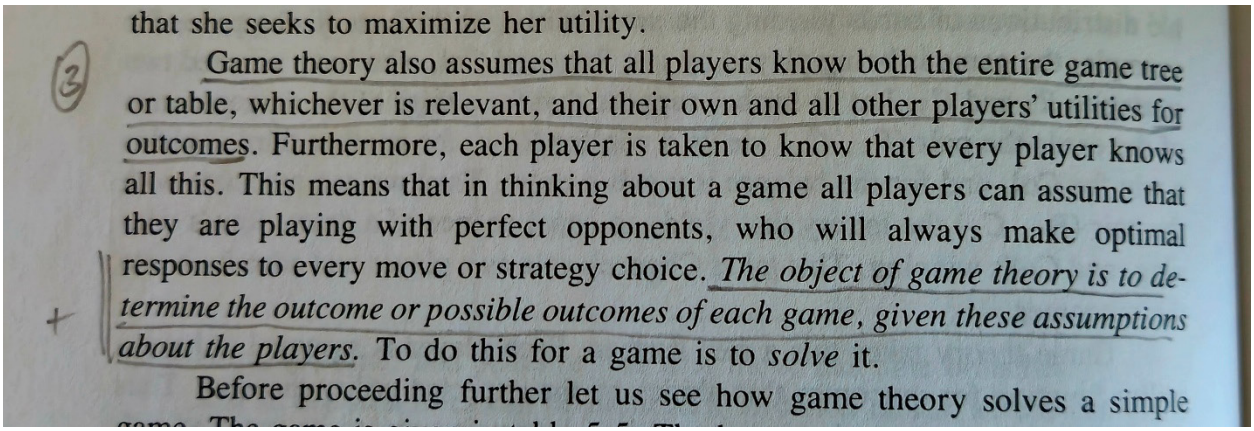
In game theory, we have two or more people (that is, *players*), although we will primarily focus on games with two players. As Resnik explains on p. 124, we still use tables. Instead of using the terms *options* or *actions*, each player has one or more *strategies*. One player's strategies are on the left, and the other player's strategies are at the top. Hence, we can call the first (left) player "Row" and the second (top) player "Column" or "Col."

		Column player		
		Col's strategy 1	**Col's strategy 2**	**Col's strategy 3**
Row player	**Row's strategy 1**	0, 0	-1, 1	1, -1
	Row's strategy 2	1, -1	0, 0	-1, 1
	Row's strategy 3	-1, 1	1, -1	0, 0

Since there are two players, each cell for the outcomes (or payoffs) has to contain an outcome for both players. The first value is for the row player and the second is for the column player. So, if Row plays strategy 1 and Col plays strategy 2, Row will get -1 and Col will get 1. (This is the gray cell.)

P. 126

Whereas we often thought about decisions in terms of what "you" should do. When analyzing a game, our goal is, generally, to determine what the two players will or should do, and then what the outcome of the game will be. To that end, the first paragraph on p. 126 is important.



that she seeks to maximize her utility.

Game theory also assumes that all players know both the entire game tree or table, whichever is relevant, and their own and all other players' utilities for outcomes. Furthermore, each player is taken to know that every player knows all this. This means that in thinking about a game all players can assume that they are playing with perfect opponents, who will always make optimal responses to every move or strategy choice. *The object of game theory is to determine the outcome or possible outcomes of each game, given these assumptions about the players.* To do this for a game is to *solve* it.

Before proceeding further let us see how game theory solves a simple

Keep in mind that (*a*) both players know all of the possible payoffs for both players, and (*b*) players want to maximize their payoffs.

Once you're used to looking at two payoffs in each cell of the game table (and knowing which one goes with which player), the next task that you have to learn is how to use the dominance principle (from decision theory) for Col. This is the game on p. 126.

	C1	C2	C3
R1	0, 0	1, 2	0, 2
R2	1, 3	1, 4	0, 0

You know how to use the dominance principle for Row. Just ignore the second value in each cell. No matter what Col does, Row does better by playing R2. (So, strategy R2 dominates strategy R1.)

	C1	C2	C3
R1	0, 0	1, 2	0, 2
R2	1, 3	1, 4	0, 0

We apply the same reasoning to Col's choices. But now we ask ourselves, 'If R1 happens, which strategy (C1, C2, or C3) is best for Col? (C2 or C3) And if R2 happens, which strategy is best for Col? (C2 again) So, C2 dominates C1 and C3.

	C1	C2	C3
R1	0, 0	1, 2	0, 2
R2	1, 3	1, 4	0, 0

So, Row will play R2 and Col will play C2. Thus, the "solution" to this game is (R2, C2).

It won't always be the case that, for both players, one strategy dominates all of the others. But, even so, we can still sometimes use the dominance principle effectively in game theory—as we will see in section 5-2.

Section 5-2

On the bottom of p. 126, Resnik distinguishes between *strictly competitive* and *non-strictly competitive* games. Sections 5-2 and 5-3, deals with the first, strictly competitive games. As Resnik explains at the beginning of 5-2,

> In a strictly competitive game the preferences of the two players are in exact opposition. To the extent one gains from an outcome of the game, the other loses. If we think of the gains and losses in terms of monetary payoffs, the winner receives the winnings in terms of a payoff from the loser.

Since the gain from winning is equivalent to the loss from losing, these games are also called *zero-sum games*. (That is, in such games, one player winning $10 means that the other player loses $10. And so, $10 + (-$10) = 0.) Since the outcomes always sum to zero, if we know one player's payoff, we also know the other player's. For this reason, Resnik only puts the payoff for the row player in each cell. I will always include both, though. Here is the complete version for the game in table 5-6 (on p. 128):

	C1	C2	C3	C4
R1	0, 0	1, -1	7, -7	7, -7
R2	4, -4	1, -1	2, -2	10, -10
R3	3, -3	1, -1	0, 0	25, -25
R4	0, 0	0, 0	7, -7	10, -10

We can solve this game using what Resnik calls "dominance reasoning"—which is also called the "iterated elimination of dominated strategies."

None of Row's strategies are dominated, but examine Col's strategies carefully.

	C1	C2	C3	C4
R1	0, 0	1, -1	7, -7	7, -7
R2	4, -4	1, -1	2, -2	10, -10
R3	3, -3	1, -1	0, 0	25, -25
R4	0, 0	0, 0	7, -7	10, -10

Between C1 and C2, is one strategy always better than the other? That is, if R1 happens, which is better? (C1) If R2 happens, which is better? (C2) If R3 happens, which is better? (C2) And if R4 happens, which is better? (C1 and C2 have the same payoffs in this case.) So, sometimes C1 does

better, sometimes C2 does better, and sometimes they do equally well. Consequently, neither strategy dominates the other.

Do the same analysis for C1 vs. C3, C1 vs. C4, C2 vs. C3, C2 vs. C4, and C3 vs. C4.

You should have found that C1, C2, and C3 all dominate C4. So, for Col, it never makes sense to choose C4. And (importantly!) Row knows that Col will never choose C4. Thus, the game is really this:

	C1	C2	C3
R1	0, 0	1, -1	7, -7
R2	4, -4	1, -1	2, -2
R3	3, -3	1, -1	0, 0
R4	0, 0	0, 0	7, -7

Now, none of Col's strategies are dominated, but R2 dominates R3.

	C1	C2	C3
R1	0, 0	1, -1	7, -7
R2	4, -4	1, -1	2, -2
R3	3, -3	1, -1	0, 0
R4	0, 0	0, 0	7, -7

This means that Row will never choose R3, and Col knows that Row will never choose R3. So, the table reduces to this:

	C1	C2	C3
R1	0, 0	1, -1	7, -7
R2	4, -4	1, -1	2, -2
R4	0, 0	0, 0	7, -7

If you examine this table, you will find that R1 dominates R4, which means this is the effective table:

	C1	C2	C3
R1	0, 0	1, -1	7, -7
R2	4, -4	1, -1	2, -2

Now, C2 dominates C3. So, we have this:

	C1	C2
R1	0, 0	1, -1
R2	4, -4	1, -1

R2 dominates R1. So, we have this:

	C1	C2
R2	4, -4	1, -1

So, Row is playing R2. For Col, C1 dominates C2. And so, we have a solution to this game:

	C2
R2	1, -1

The solution is (R2, C2).